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# **Search for axionlike particle as dark matter candidate with nuclear magnetic resonance**

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# Abstract

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Axions and axionlike particles (ALPs) are theoretically well-motivated candidates for the dark matter that constitutes approximately 27 % of the total energy content of the universe. Through the nuclear gradient coupling, an oscillating ALP dark-matter field acts as an effective pseudomagnetic field on nuclear spins, driving resonant spin precession at the ALP Compton frequency. Nuclear magnetic resonance (NMR) is among the most sensitive techniques for detecting weak oscillating fields, making it a natural tool for direct ALP dark-matter searches.

This thesis presents a theoretical and experimental investigation of ALP dark-matter searches using precision NMR in the context of the Cosmic Axion Spin Precession Experiment (CASPER). On the theoretical side, the signal amplitude and optimal measurement strategy for a frequency-scanning search are derived analytically. Depending on the relative magnitudes of the NMR coherence time  $T_2^*$ , the intrinsic transverse relaxation time  $T_2$ , and the ALP coherence time  $\tau_a$ , four distinct signal regimes are identified, each with a characteristic sensitivity and optimal dwell-time allocation. In all regimes, the sensitivity to the ALP-nucleon coupling strength scales as  $T_{\text{tot}}^{-1/4}$  with total measurement time, motivating improvements in spin polarization and relaxation times over simply accumulating more data. The optimal frequency step is equal to the sensitive bandwidth of a single measurement, and the dwell time should be distributed uniformly across frequency steps in regimes (1)–(3), or proportional to  $\nu^{-1}$  in regime (4).

On the experimental side, the scanning framework is applied in a proof-of-concept search for the ALP-proton gradient coupling using the CASPER-Gradient-Low-Field apparatus at the Johannes Gutenberg University Mainz. A thermally polarized liquid methanol sample is used to scan a 240 Hz band centred at 1.348570 MHz (proton Larmor frequency), corresponding to ALP masses in the range 5.576741–5.577733 neV/ $c^2$ . Three sequential scans were performed; no ALP signal candidates survived a cross-scan consistency check. A 95 % confidence-level upper limit of  $|g_{\text{ap}}| \lesssim 3 \times 10^{-2} \text{ GeV}^{-1}$  is set on the ALP-proton gradient coupling in this mass range, constituting the first dedicated laboratory search therein.

The sensitivity of the current measurement is dominated by the small thermal polarization of the methanol sample ( $\approx 1.8 \times 10^{-7}$ ). Projected sensitivity estimates demonstrate that hyperpolarized samples—parahydrogen-induced polarization, dynamic nuclear polarization, or spin-exchange optical pumping of liquid

$^{129}\text{Xe}$ —can improve the coupling sensitivity by several orders of magnitude, opening a path toward new physics beyond current bounds.

# Acknowledgments

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The central observation motivating this thesis is that axionlike particles (ALPs) interact with nuclear spins as an effective oscillating pseudomagnetic field. Nuclear magnetic resonance (NMR) is among the most sensitive techniques for detecting weak oscillating fields, making it a natural tool for direct ALP dark-matter searches.

Astrophysical evidence for dark matter comes from galaxy rotation curves, gravitational lensing, large-scale structure, and the cosmic microwave background: visible matter alone cannot account for the observed dynamics. Axions and axionlike particles are theoretically attractive candidates because they are weakly coupled, naturally light, and can generate coherent spin-precession signals accessible to precision NMR experiments.

This chapter introduces the key ingredients of the search. Section 1.1 reviews the NMR concepts most relevant to the experiment. Section 1.2 summarises the dark-matter evidence. Section 1.3 describes the axion and ALP framework, including the pseudomagnetic-field analogy that connects ALP physics to NMR. Section 1.4 outlines the CASPEr experimental programme. The theory of signal estimation and scanning optimisation is developed in Chapter 2; the apparatus is described in Chapter 3; measurements and results appear in Chapter 4.

## 1.1 Nuclear magnetic resonance

Nuclear magnetic resonance is a technique for studying nuclear spins in magnetic fields. In the presence of a leading field, spin states split by the Zeeman effect and a polarized sample acquires a net magnetization that precesses at the Larmor frequency. Because the resonant frequency is directly proportional to the magnetic field, NMR is a precision tool for field measurement.

For the purposes of this thesis, the most important features of NMR are its sensitivity to weak fields, its applicability from zero to very high field, and its rich time-domain response. Those properties make it possible to reinterpret axionlike spin precession as a small perturbation on an otherwise standard NMR experiment.

### 1.1.1 Zeeman splitting and polarization

In a magnetic field  $B_0$ , a spin-1/2 nucleus splits into two energy levels separated by

$$\Delta E = \hbar \gamma B_0, \quad (1.1)$$

where  $\gamma$  is the nuclear gyromagnetic ratio. This energy difference sets the Larmor (resonance) frequency

$$\nu_L = \frac{\gamma B_0}{2\pi}. \quad (1.2)$$

For protons,  $\gamma/(2\pi) = 42.577 \text{ MHz/T}$ , so a bias field of 31.67 mT (as used in this work) gives  $\nu_L \approx 1.348 \text{ MHz}$ .

At thermal equilibrium, the spin populations obey the Boltzmann distribution. In the high-temperature limit  $\Delta E \ll k_B T$  (satisfied for all conditions in this thesis), the fractional spin polarization is

$$p \approx \frac{\hbar \gamma B_0}{2k_B T}. \quad (1.3)$$

At  $B_0 = 31.67 \text{ mT}$  and  $T = 180 \text{ K}$ , the proton thermal polarization is only  $p \approx 1.8 \times 10^{-7}$ . Hyperpolarization techniques—such as spin-exchange optical pumping or parahydrogen-induced polarization—can boost  $p$  by several orders of magnitude, which is the main route to improved sensitivity identified in this thesis.

### 1.1.2 Spin-projection noise

Even in the absence of a leading field, an ensemble of  $N$  spins does not cancel perfectly. Statistical fluctuations leave a residual transverse moment of order  $\sqrt{N} \mu$  (where  $\mu = \hbar \gamma / 2$  is the nuclear magneton), giving a magnetization noise floor

$$M_{\text{SPN}} \sim \frac{\sqrt{N} \mu}{V}. \quad (1.4)$$

This spin-projection noise (SPN) is the fundamental quantum limit for spin measurements [18]. In the axion-search context it sets the minimum detectable transverse magnetization and is listed in Table 3.2 for the experimental parameters of this work.

### 1.1.3 Relaxations

Relaxation describes the return of the spin ensemble toward equilibrium after it has been perturbed. Longitudinal relaxation  $T_1$  controls how quickly the magnetization along the leading field rebuilds, while transverse relaxation determines how quickly coherent precession dephases. Two transverse relaxation times must be distinguished:

$$\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T_\delta}, \quad (1.5)$$

where  $T_2$  characterises irreversible spin-spin interactions and  $T_\delta$  quantifies the additional dephasing caused by magnetic-field inhomogeneity across the sample volume. Because field inhomogeneity can be reduced by shimming (Chapter 3),  $T_2^*$  is experimentally controllable, but  $T_2$  sets an intrinsic upper bound.

These time scales are central to the thesis because they set the window over which the ALP-induced signal can be observed coherently. Longer  $T_2^*$  improves sensitivity in all four signal regimes derived in Chapter 2.

#### $T_1$ relaxation

Longitudinal relaxation governs how the population difference between spin states rebuilds after perturbation. In practice, a short  $T_1$  is helpful for rapid repetition, while a long  $T_1$  can preserve polarization for longer measurements.

#### $T_2^*$ relaxation

Transverse relaxation combines microscopic spin-spin interactions ( $T_2$ ) and macroscopic field inhomogeneity ( $T_\delta$ ) as given in Equation (1.5). It determines the decay of the free-induction signal and directly controls the linewidth in the frequency domain. For field searches, a narrower linewidth can improve sensitivity, but only if the coherence time remains long enough to integrate signal power efficiently.

#### Cross relaxation

Cross relaxation is one of the mechanisms by which energy can be exchanged between coupled spin species or spin reservoirs. It is relevant whenever the sample or environment contains more than one spin system, because it can alter the effective relaxation rates seen in the measurement.

### 1.1.4 Chemical shift

Chemical shielding changes the local magnetic field felt by a nucleus and therefore shifts the resonance frequency. The shift is typically small on an absolute scale, but it can be crucial in spectroscopy because it makes nuclei in different chemical environments distinguishable.

In the current thesis, chemical shift matters mainly as a source of spectral structure and as a reminder that the resonant response is not determined by the external field alone.

### 1.1.5 Spin-spin coupling

Coupling between nuclear spins splits resonance lines into multiplets. This effect is useful rather than harmful when the goal is signal verification, because the pattern of peaks can identify a particular molecular species and help distinguish genuine NMR signals from noise spikes.

For the calibration sample used later in the thesis, J-coupling provides a clear spectral signature that is much easier to recognize than a single unresolved line.

### 1.1.6 Experimental schemes

Two pulse-based schemes are especially useful. Free induction decay measures the coherent response after a tipping pulse, while spin echo refocuses dephasing and gives access to the intrinsic transverse relaxation time. Both are diagnostic tools that help characterize the apparatus before an axion measurement.

#### Free decay

After a  $\pi/2$  pulse, the magnetization is tipped into the transverse plane and then decays with the dephasing time  $T_2^*$ . The resulting free-induction signal is the simplest way to measure resonance frequency and linewidth.

#### Spin echo

Spin echo uses a  $\pi$  pulse to reverse dephasing from static field inhomogeneity. By recovering the signal at the echo time, it separates true spin-spin relaxation from broadening caused by imperfect field homogeneity, yielding a direct measurement of  $T_2$ .

## 1.2 Dark matter hypothesis

The concept of dark matter arose from astronomical observations that could not be explained by visible matter and standard gravity alone. Galaxy rotation curves are the classic example: the outer parts of galaxies rotate too quickly for the observed baryonic mass to provide the required gravitational binding [22, 23]. The Bullet Cluster provides direct evidence through the spatial separation between the gravitational mass (traced by lensing) and the baryonic mass (X-ray gas) after two galaxy clusters collide [10]. The cosmic microwave background power spectrum and big-bang nucleosynthesis abundances independently constrain baryonic matter to roughly 5 % of the total energy budget, with dark matter contributing about 27 % [3].

### 1.2.1 History

From early galaxy-dynamics studies to later lensing and large-scale structure measurements, the dark-matter hypothesis was strengthened by evidence from many independent observations. The basic conclusion is that a large fraction of the matter content of the universe is not directly visible.

### 1.2.2 Recent developments

Modern dark-matter searches span weakly interacting massive particles (WIMPs), ultralight bosons, and direct-detection strategies based on precision measurement [16]. WIMPs have been strongly constrained by direct-detection experiments; axions and axionlike particles have attracted increasing attention because they connect cosmology with laboratory-scale experiments through tiny but coherent effects.

## 1.3 Axions and axionlike particles

The QCD axion was originally proposed in 1977 by Peccei and Quinn as a solution to the strong  $CP$  problem [19, 20]. A global symmetry breaking at the scale  $f_a$  generates a pseudo-Goldstone boson whose potential is minimised at the  $CP$ -conserving point, naturally explaining why the strong interaction conserves  $CP$ . The axion acquires a mass  $m_a \sim \Lambda_{\text{QCD}}^2/f_a$  from QCD dynamics.

### 1.3.1 QCD axion

The QCD axion is a hypothetical pseudoscalar particle that appears when a global symmetry is introduced to explain why strong interactions conserve  $CP$  so well. Its properties are tightly linked to the QCD scale, which makes it a special case rather than the most general one. Cosmological production via the misalignment mechanism constrains  $f_a \lesssim 10^{12}$  GeV (axion mass  $\gtrsim 6 \mu\text{eV}$ ) if the axion is to constitute all dark matter [1, 11, 21].

### 1.3.2 Axionlike particles

Axionlike particles share the oscillatory and weakly coupled character of the QCD axion, but their mass range can be much wider. This flexibility makes them attractive targets for laboratory searches.

On time scales shorter than the ALP coherence time, the ALP field can be treated as a classical oscillation:

$$a(\mathbf{r}, t) = a_0 \cos(v_a t - \mathbf{k} \cdot \mathbf{r} + \phi), \quad (1.6)$$

where  $v_a = m_a c^2 / \hbar$  is the ALP Compton frequency,  $\mathbf{k} = m_a \mathbf{v}_a / \hbar$  is the wave vector,  $a_0$  is the amplitude, and  $\phi$  is a random phase. The amplitude  $a_0$  is set by the local dark-matter density through  $\rho_{DM} \approx c m_a^2 a_0^2 / (2\hbar^3)$  [9]. Because the ALP number density is so high for ultralight masses, this classical-field description is appropriate.

The virialized ALP field is stochastic: individual ALP quanta have a spread of velocities following the Maxwell-Boltzmann distribution with dispersion  $v_a \approx 220 \text{ km s}^{-1}$  [12]. The resulting spectral width of the ALP field sets a coherence time

$$\tau_a \approx \frac{Q_a}{v_a}, \quad Q_a \equiv \left( \frac{c}{v_a} \right)^2 \approx 1.8 \times 10^6. \quad (1.7)$$

For Compton frequencies above 1 kHz,  $\tau_a \lesssim 1000$  s. The quality factor  $Q_a$  is analogous to that of a driven oscillator: within one coherence time the ALP field behaves as a coherent monochromatic wave, but its phase randomises on longer time scales.

### 1.3.3 Non-gravitational interactions

The most relevant non-gravitational couplings for this thesis are the gradient coupling to nucleons. The nucleon Hamiltonian in the presence of an ALP field back-

ground is [14]

$$H_{\text{aNN}} \sim g_{\text{aNN}} \nabla a \cdot \mathbf{I}, \quad (1.8)$$

where  $g_{\text{aNN}}$  is the coupling strength in  $\text{GeV}^{-1}$  and  $\mathbf{I}$  is the nuclear spin operator. Comparing this with the NMR Hamiltonian  $H_{\text{NMR}} = \hbar \gamma \mathbf{B} \cdot \mathbf{I}$ , the ALP gradient acts as an effective oscillating pseudomagnetic field

$$\mathbf{B}_a(t) = \frac{g_{\text{aNN}} \nabla a}{\gamma}. \quad (1.9)$$

This gradient coupling oscillates at the ALP Compton frequency and can drive resonant spin precession when the Larmor frequency matches  $\nu_a$ . The search for this coupling is the subject of Chapter 4.

## 1.4 CASPER project

CASPER (Cosmic Axion Spin Precession Experiment) is a family of experiments proposed in 2014 [7] that use nuclear spins to search for axion-induced effects. The idea is to translate the tiny axion coupling in Equation (1.9) into a measurable spin precession in a carefully controlled NMR setup.

The project has several branches, including electric and gradient-based searches, and both low-field and high-field implementations. The low-field regime is especially relevant when the goal is to scan a wide frequency band with high sensitivity to weak couplings.

### 1.4.1 Idea

The central idea is to tune the NMR resonance to a frequency that matches the axion Compton frequency. If the two frequencies align, the axion field acts as the pseudomagnetic field  $\mathbf{B}_a(t)$  in Equation (1.9) and induces a coherent transverse magnetization in the sample.

In practice, the method reduces to a frequency scan: either the signal appears at a particular resonance or the absence of a signal constrains the coupling strength over the scanned band.

### 1.4.2 CASPER-Electric

CASPER-Electric (CASPER-E) searches for an oscillating nuclear electric dipole moment induced by the ALP-gluon coupling. This moment generates an effective

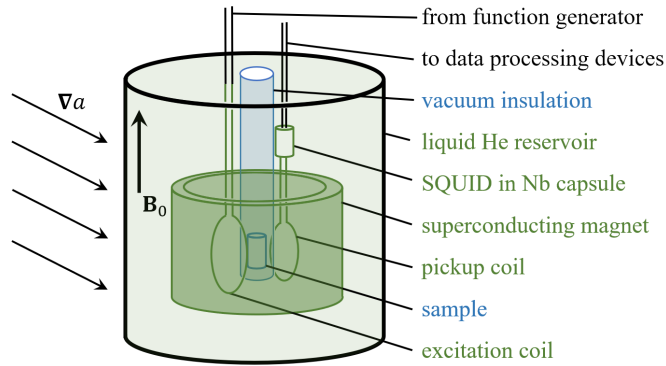
oscillating field along the nuclear spin axis in the presence of an electric field, enabling detection through NMR at the corresponding Compton frequency.

### 1.4.3 CASPEr comagnetometers

Comagnetometer configurations use two nuclear spin species in the same volume to suppress common-mode noise from magnetic-field fluctuations. The differential signal between the two species enhances sensitivity to a pseudomagnetic ALP field while rejecting electromagnetic backgrounds. Early results were obtained with a liquid-state comagnetometer [26] and an ultralow-field configuration [13].

### 1.4.4 CASPEr-Gradient

CASPEr-Gradient searches for the ALP-fermion gradient coupling described by Equation (2.5). Several setups at Mainz cover different ALP-mass ranges. The CASPEr-Gradient-Low-Field experiment uses bias fields up to about 0.1 T, corresponding to proton Larmor frequencies up to approximately 4.3 MHz. The CASPEr-Gradient-High-Field experiment will use a tunable 14.1 T magnet to reach frequencies up to 600 MHz. A conceptual diagram of the low-field apparatus is shown in Figure 1.1. The cryostat contains a liquid helium reservoir, excitation and pickup coils, and a SQUID magnetometer. A superconducting magnet produces the bias field  $B_0$ , which sets the Larmor frequency and can be varied to scan ALP masses.



**Figure 1.1:** Conceptual diagram of the CASPEr-Gradient-Low-Field setup. Superconducting devices (dark green) are submerged in a liquid helium bath (light green). The excitation and pickup coils characterise the sample and detect the spin-precession signal. The sample temperature can be controlled independently of the helium bath, enabling measurements with both thermal and hyperpolarized samples. Reproduced from Ref. [28].

*Part of the content in this chapter related to frequency-scanning considerations is adapted from Ref. [28].*

The ALP field can be detected through its gradient coupling to nuclear spins, which acts as an effective oscillating pseudomagnetic field. Understanding the properties of this coupling—its frequency, amplitude, and coherence—is essential for designing a sensitive experiment and choosing an optimal measurement strategy. By answering the questions of what the signal looks like and how it depends on experimental parameters, this chapter lays the groundwork for the scanning strategy and sensitivity analysis of Chapter 4.

## 2.1 Stochasticity

On time scales shorter than its coherence time  $\tau_a$ , the ALP field can be approximated as a coherent classical oscillation:

$$a(\mathbf{r}, t) = a_0 \cos(\nu_a t - \mathbf{k} \cdot \mathbf{r} + \phi). \quad (2.1)$$

Here  $a_0$  is the amplitude of the field,  $\nu_a = m_a c^2 / \hbar$  is the ALP Compton frequency where  $c$  is the speed of light and  $\hbar$  is the reduced Planck constant,  $\mathbf{k} = m_a \mathbf{v}_a / \hbar$  is the wave vector ( $\mathbf{v}_a$  is the relative velocity of the ALP field and the detector),  $\mathbf{r}$  is the displacement vector, and  $\phi$  is a random phase uniformly distributed in  $[0, 2\pi)$ . The amplitude  $a_0$  follows a Rayleigh distribution. Its average root-mean-square (r.m.s.) value is set by the local dark-matter density  $\rho_{DM} \approx 0.4 \text{ GeVcm}^{-3}$  [6, 17, 23]:

$$\rho_{DM} \approx \frac{c m_a^2 a_0^2}{2\hbar^3}. \quad (2.2)$$

The ALP field is stochastic in nature [9]. Assuming that ALP dark matter is virialized in the galactic potential and follows a Maxwell-Boltzmann velocity distribution with average speed  $v_a \approx 220 \text{ km s}^{-1}$  [12], the spectral linewidth of the ALP field observed with a detector on Earth is

$$\Gamma_a = \frac{\nu_a}{Q_a}, \quad (2.3)$$

where  $Q_a \equiv (c/v_a)^2 \approx 1.8 \times 10^6$  is the ALP quality factor. The ALP coherence time  $\tau_a$  is related to the linewidth by  $\Gamma_a = 2\pi/\tau_a$ , giving

$$\tau_a = \frac{2\pi Q_a}{\nu_a}. \quad (2.4)$$

At any given search frequency,  $\tau_a$  is fixed by the ALP mass and the dark-matter velocity model. For Compton frequencies above 1 kHz,  $\tau_a \lesssim 1000$  s.

## 2.2 Periodic modulations

### 2.2.1 Daily modulation

The ALP-field gradient driving spin precession depends on the direction of the relative velocity between the Earth and the local dark-matter halo. As the Earth rotates, this direction sweeps through space with a period of one sidereal day. An ALP signal therefore exhibits a daily modulation of its amplitude and phase [15].

### 2.2.2 Annual modulation

In addition to daily rotation, the Earth's orbital velocity around the Sun modulates the relative ALP-detector velocity with a period of one year. Both the daily and annual modulations have precisely predictable periods and phases, making them useful consistency checks for any claimed dark-matter-candidate signal [15].

## 2.3 Spin-precession signal

The nuclear-gradient coupling of the ALP field to nuclear spins is described by the Hamiltonian

$$H_{\text{aNN}} \sim g_{\text{aNN}} \nabla a \cdot \mathbf{I}, \quad (2.5)$$

where  $g_{\text{aNN}}$  is the coupling strength in units of  $\text{GeV}^{-1}$ ,  $\nabla a$  is the gradient of the ALP field and  $\mathbf{I}$  is the nuclear spin operator. This coupling acts as a pseudomagnetic drive field for nuclear spins precessing at the Larmor frequency in a bias field  $\mathbf{B}_0$ . Its r.m.s. Rabi frequency is [9, 28]

$$\Omega_a \approx \frac{1}{2} g_{\text{aNN}} \sqrt{2\hbar c \rho_{DM} v_{\perp}}, \quad (2.6)$$

where  $v_{\perp}$  is the component of the ALP-detector relative velocity perpendicular to  $\mathbf{B}_0$ . Taking  $v_{\perp} = 220 \text{ km s}^{-1}$ , we have  $\Omega_a/2\pi = g_{a\text{NN}} \times 0.2 \text{ GeV} \cdot \text{Hz}$ . For  $g_{a\text{NN}} \lesssim 10^{-10} \text{ GeV}^{-1}$ , the Rabi period  $2\pi/\Omega_a \gtrsim 1000 \text{ yr}$ , confirming that the ALP field is a weak drive on the spins.

### 2.3.1 Relaxations and coherence

The transverse spin relaxation in the sample arises from homogeneous mechanisms (e.g., spin-spin interactions), characterised by the time  $T_2$ , and from inhomogeneous mechanisms (e.g., magnetic-field gradients across the sample), which cause additional dephasing. The total effective transverse relaxation time  $T_2^* \leq T_2$  incorporates both contributions.

The NMR linewidth (FWHM, assuming a Lorentzian lineshape) of the free-induction-decay spectrum is

$$\Gamma_n = \frac{2}{T_2^*}, \quad (2.7)$$

and the normalized linewidth is

$$\delta_v \equiv \frac{\Gamma_n}{\nu} = \frac{2}{\nu T_2^*}, \quad (2.8)$$

often expressed in parts per million (ppm). The inhomogeneous contribution to  $1/T_2^*$  can be reduced by magnetic shimming, giving control over  $T_2^*$  independently of the intrinsic  $T_2$ .

### 2.3.2 Estimation

For a sample with equilibrium longitudinal magnetization  $M_0$  prepared along  $\mathbf{B}_0$ , the ALP-field drive tilts the magnetization by a small angle  $\xi \ll 1$  (weak-drive regime,  $\Omega_a T \ll 1$ ), generating a transverse magnetization  $M_1 \approx M_0 \xi$ . The r.m.s. transverse magnetization is [28]

$$\sqrt{\langle M_1^2 \rangle} \approx \frac{1}{\sqrt{2}} u_n M_0 \sqrt{\langle \xi^2 \rangle}, \quad (2.9)$$

where  $u_n$  is the fraction of nuclear spins on resonance with the ALP field. Approximating both the NMR and ALP lineshapes as rectangles,

$$u_n \approx \begin{cases} 1, & \tau_a \ll T_2^* \\ \frac{\Gamma_a}{\Gamma_n} = \frac{\pi T_2^*}{\tau_a}, & T_2^* \ll \tau_a \end{cases}. \quad (2.10)$$

The r.m.s. tipping angle  $\sqrt{\langle \xi^2 \rangle}$  is limited by the interplay between ALP decoherence and spin relaxation. Modeling the ALP decoherence as a random phase that resets on the time scale  $\tau_a$ , the tipping angle executes a two-dimensional random walk in the rotating frame with step size  $\Omega_a \tau_a$  [28]. Combining this with the three relevant relaxation scales,

$$\sqrt{\langle \xi^2 \rangle} = \begin{cases} \Omega_a \sqrt{\tau_a T_2^*}, & \tau_a \ll T_2^* \\ \Omega_a \frac{\tau_a}{\sqrt{\pi}}, & T_2^* \ll \tau_a \ll T_2 \\ \Omega_a T_2, & T_2 \ll \tau_a \end{cases}. \quad (2.11)$$

These cases combine into the r.m.s. transverse magnetization (for dwell time  $T \gg \tau_a, T_2^*, T_2$ ):

$$\sqrt{\langle M_1^2 \rangle} \approx \frac{M_0 \Omega_a}{\sqrt{2}} \begin{cases} \sqrt{\tau_a T_2^*}, & \tau_a \ll T_2^* \\ \frac{T_2^*}{\sqrt{\pi}}, & T_2^* \ll \tau_a \ll T_2 \\ \frac{T_2^* T_2}{\sqrt{\pi} \tau_a}, & T_2 \ll \tau_a \end{cases}. \quad (2.12)$$

In the third case, when  $T_2 \ll \tau_a$  and the dwell time  $T \gg \tau_a$ , the signal amplitude decreases because the ALP field cannot drive the spins coherently for longer than  $T_2$ , while off-resonance spins (those not within the narrow ALP linewidth) contribute negligibly.

### 2.3.3 Numerical simulation

The analytical estimates above are valid in the limit  $T \gg \tau_a, T_2^*, T_2$ . At lower Compton frequencies, the ALP coherence time can become comparable to or longer than experimentally accessible dwell times, and the simple scaling laws no longer ap-

ply. Numerical simulations using the `axionbloch` package are used to validate the analytical estimates and to explore intermediate parameter regimes.

Three representative situations arise:

**Low frequency**  $\tau_a \gg T$ : the ALP field is essentially coherent over the measurement. The signal grows linearly in time and the  $T^{-1/4}$  sensitivity scaling is not yet reached.

**Medium frequency**  $\tau_a \approx T$ : the transition regime, treated numerically.

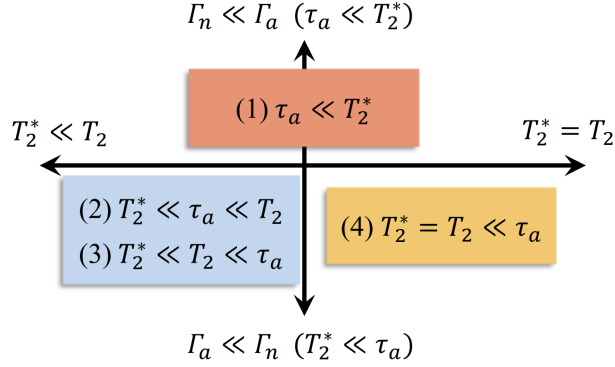
**High frequency**  $T \gg \tau_a, T_2^*, T_2$ : the analytical results apply. For hyperpolarized  $^{129}\text{Xe}$ ,  $T_1$  relaxation limits the achievable polarization, introducing an additional reduction factor of approximately 1/3 relative to the fully polarized case.

### 2.3.4 Summary

Depending on the relative magnitudes of  $T_2$ ,  $T_2^*$ , and  $\tau_a$ , four regimes are identified [28]:

- (1)  $\tau_a \ll T_2^*$ ;
- (2)  $T_2^* \ll \tau_a \ll T_2$ ;
- (3)  $T_2^* \ll T_2 \ll \tau_a$ ;
- (4)  $T_2^* = T_2 \ll \tau_a$ .

These regimes are illustrated in Figure 2.1. The key parameters for each regime are summarised in Table 2.1, where  $\nu_{\text{start}}$  and  $\nu_{\text{end}}$  are the lower and upper limits of the target frequency band.



**Figure 2.1:** Regimes of the spin-precession experiment based on the relative magnitudes of  $T_2$ ,  $T_2^*$ , and  $\tau_a$ . Reproduced from Ref. [28].

**Table 2.1:** Summary of parameters for the four regimes. The normalised linewidth  $\delta_\nu = \tau_a/(\pi Q_a T_2^*)$  is assumed constant in each regime. The optimal dwell time is derived in Appendix C.

|     | Regime                     | Optimal dwell time $T$                                                                 | Signal linewidth $\Gamma$ | Spectral factor $u_n$      | r.m.s. tip |
|-----|----------------------------|----------------------------------------------------------------------------------------|---------------------------|----------------------------|------------|
| (1) | $\tau_a \ll T_2^*$         | $\frac{T_{\text{tot}}}{Q_a \ln(\nu_{\text{end}}/\nu_{\text{start}})}$                  | $\frac{2}{T_2^*}$         | 1                          |            |
| (2) | $T_2^* \ll \tau_a \ll T_2$ | $\frac{T_{\text{tot}} \tau_a}{\pi Q_a T_2^* \ln(\nu_{\text{end}}/\nu_{\text{start}})}$ | $\frac{2\pi}{\tau_a}$     | $\frac{\pi T_2^*}{\tau_a}$ |            |
| (3) | $T_2^* \ll T_2 \ll \tau_a$ | $\frac{T_{\text{tot}} \tau_a}{\pi Q_a T_2^* \ln(\nu_{\text{end}}/\nu_{\text{start}})}$ | $\frac{2\pi}{\tau_a}$     | $\frac{\pi T_2^*}{\tau_a}$ |            |
| (4) | $T_2^* = T_2 \ll \tau_a$   | $\frac{T_{\text{tot}} \tau_a}{\pi Q_a T_2 \ln(\nu_{\text{end}}/\nu_{\text{start}})}$   | $\frac{2\pi}{\tau_a}$     | $\frac{\pi T_2}{\tau_a}$   |            |

*The content of this chapter is adapted from Ref. [25].*

The CASPEr-Gradient-Low-Field apparatus converts a nuclear-spin signal into a detectable electrical output through three functional blocks: the magnet system sets the operating bias field and thereby the Larmor frequency of the sample; the sample and pickup coil generate and capture the magnetic response; and the SQUID and lock-in amplifier chain convert, amplify, and record the signal. This chapter describes each block in turn and closes with a summary of the key parameters.

### 3.1 Magnet system

The apparatus is built around a flow-through liquid-helium cryostat housing a superconducting solenoid that can generate a bias field  $B_0$  up to 0.1 T. The DM mass range accessible to the experiment depends on the gyromagnetic ratio of the sample and on  $B_0$ ; for proton spins, the covered Larmor frequency range is approximately 1 kHz to 4.3 MHz.

Eight superconducting shim coils surrounding the solenoid are used to reduce the field inhomogeneity to less than 2 ppm over a spherical volume with radius 4 mm. In the measurements reported here, an inhomogeneity of approximately 10 ppm was achieved after tuning the shim fields using a Bayesian-optimisation algorithm [24]. The larger-than-design inhomogeneity results from using a longer sample (24 mm height) and from the insertion of an additional superconducting pickup coil.

Helmholtz coils wound outside the cryostat provide a controllable offset on  $B_0$ , enabling fine-tuning of the Larmor frequency without ramping the main solenoid. This arrangement is used to step the Larmor frequency in 6 Hz increments during the scan.

## 3.2 Sample

Liquid methanol ( $\text{CH}_3\text{OH}$ ) was chosen as the sample because of its high proton density ( $99 \text{ mmol/cm}^3$ ) and low freezing point ( $175.6 \text{ K}$ ), which allows it to remain liquid at the operating temperature of approximately  $180 \text{ K}$ . With four protons per molecule, the  $1.2 \text{ cm}^3$  cylindrical sample (radius  $4 \text{ mm}$ , height  $24 \text{ mm}$ ) contains  $7.2 \times 10^{22}$  protons.

A variable-temperature insert (VTI) positions the sample in the homogeneous region of the bias field and thermally isolates it from the liquid-helium bath. The sample temperature is regulated by flowing warm nitrogen gas through a spiral surrounding the sample holder. Temperature stability is monitored by repeated pulsed-NMR measurements: a constant NMR linewidth indicates a stable temperature.

At thermal equilibrium in the bias field  $B_0$ , the proton spins acquire a net longitudinal magnetization

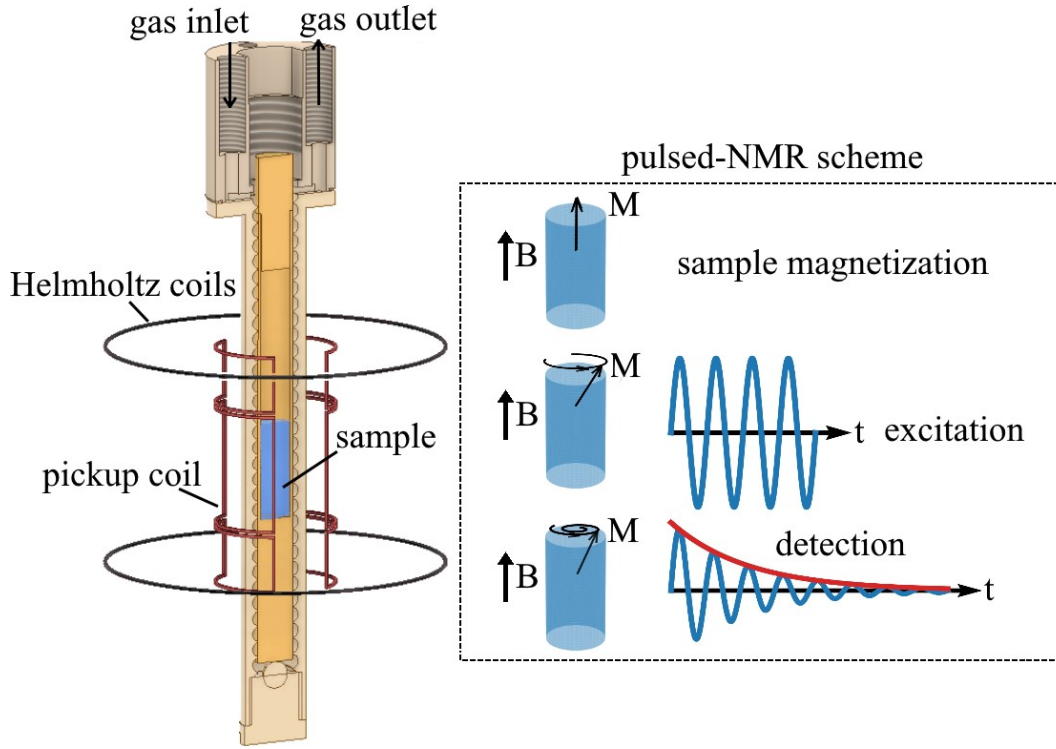
$$M_0 = \frac{n \gamma^2 \hbar^2 I(I+1) B_0}{3 k_B T}, \quad (3.1)$$

where  $n$  is the proton number density,  $\gamma = 2\pi \times 42.577 \text{ MHz/T}$  is the proton gyromagnetic ratio,  $I = 1/2$ ,  $k_B$  is the Boltzmann constant, and  $T$  is the sample temperature. At  $B_0 = 31.67 \text{ mT}$  and  $T \approx 180 \text{ K}$ , the thermal polarization is  $\approx 1.8 \times 10^{-7}$  and  $M_0 \approx 1.86 \times 10^{-10} \text{ T}/\mu_0$ .

## 3.3 Pickup coil and shielding

A niobium (Nb) pickup coil mounted on the outside of the VTI in the liquid-helium bath detects the transverse magnetization of the sample. The coil consists of three saddle-pair coils in a gradiometer configuration: the area of the central coil is twice that of the two outer coils, and the coils are wound in opposing senses so that far-field flux contributions cancel. The filling factor—the fraction of the coil volume occupied by the sample—is approximately  $8 \%$ .

A superconducting Nb shield built into the cryostat attenuates low-frequency magnetic noise. Additional shielding is provided by a  $\mu$ -metal shield outside the cryogenic reservoir and by a Faraday cage enclosing the entire apparatus, which suppresses electromagnetic interference (EMI). The pickup coil geometry and pulsed-NMR scheme are illustrated in Figure 3.1.



**Figure 3.1:** Left: the methanol sample (blue) sits at the centre of the three-coil Nb saddle-pair pickup (red). Helmholtz coils (black) fine-tune the bias field. Right: pulsed-NMR scheme used for sample characterisation. The net magnetization  $\mathbf{M}$  is tipped into the transverse plane by a  $\pi/2$  pulse and precesses, generating a signal in the pickup and SQUID. Reproduced from Ref. [25].

### 3.4 Detection chain

Magnetic flux in the pickup coil is inductively coupled into a direct-current superconducting quantum interference device (dc-SQUID) operated in flux-locked-loop (FLL) mode. The properties of the SQUID sensor are listed in Table 3.1. The feedback voltage  $V_f$  is related to the magnetic flux in the pickup coil  $\Phi_{\text{pick}}$  by

$$V_f = \frac{R_f}{M_f} \frac{M_{\text{in}}}{L_{\text{pick}} + L_{\text{in}}} \Phi_{\text{pick}} = \varepsilon \Phi_{\text{pick}}, \quad (3.2)$$

where  $R_f$  is the feedback resistance,  $M_f$  is the feedback-coil mutual inductance,  $M_{\text{in}}$  is the mutual inductance between pickup and SQUID input coils, and  $L_{\text{pick}}$ ,  $L_{\text{in}}$  are

the respective coil inductances. The conversion factor  $\varepsilon \approx 0.549 \text{ mV}/\Phi_0$ , where  $\Phi_0$  is the magnetic flux quantum.

**Table 3.1:** Properties of the dc-SQUID sensor used in the measurements.

|                                                 |                          |
|-------------------------------------------------|--------------------------|
| Pickup coil inductance ( $L_{\text{pick}}$ )    | 553 nH                   |
| SQUID input coil inductance ( $L_{\text{in}}$ ) | 400 nH                   |
| Mutual inductance ( $M_{\text{in}}$ )           | 3.98 nH                  |
| Feedback-coil inductance ( $M_{\text{f}}$ )     | $22.83 \Phi_0/\text{mA}$ |
| Feedback resistance ( $R_{\text{f}}$ )          | 3 k $\Omega$             |

The detection chain is characterised by the transfer coefficient  $\beta$ , defined as the ratio of the SQUID feedback-voltage amplitude  $V_{\perp}$  to the transverse sample magnetization  $\mu_0 M_{\perp}$ :

$$\beta = \frac{V_{\perp}}{\mu_0 M_{\perp}}. \quad (3.3)$$

The transfer coefficient was determined at each scan step from the feedback voltage following a  $\pi/2$  pulse, for which  $M_{\perp} = M_0$ . The average value is  $\beta \approx 4 \times 10^6 \text{ V/T}$ .

During ALP-search measurements the SQUID and FLL electronics are battery-powered to avoid electromagnetic interference from the mains. All other electronics (lock-in amplifier, field-ramp supply) are outside the Faraday cage; only shielded BNC cables penetrating the cage are used to carry signals.

The FLL output is fed into a Zurich Instruments MFLI Lock-in Amplifier (LIA), which demodulates the signal into in-phase (I) and quadrature (Q) channels. The complex output  $I + iQ$  is recorded at a sampling frequency of 13.4 kHz and is used to generate power spectral densities (PSDs) via fast Fourier transformation (FFT).

### 3.5 Parameters summary

Table 3.2 summarises the key parameters of the CASPER-Gradient-LF apparatus for the methanol measurement reported in this thesis and for two projected future configurations using hyperpolarized samples. The thermal polarization of methanol is the dominant limiting factor in the current measurement. Hyperpo-

larized methanol or liquid  $^{129}\text{Xe}$  could boost the polarization by several orders of magnitude, substantially improving the sensitivity.

**Table 3.2:** Parameters of the CASPER-Gradient-LF apparatus for the current measurement (thermally polarized methanol) and two projected future configurations. Polarization is given as molar polarization (polarization $\times$ concentration).

|                                                     |  | This work                       | Examples of future CASPER-Gradient-LF                  |                                       |
|-----------------------------------------------------|--|---------------------------------|--------------------------------------------------------|---------------------------------------|
| Sample                                              |  | Methanol                        | Methanol <sup>a</sup>                                  | Liquid <sup>129</sup> Xe <sup>b</sup> |
| Spin density (cm <sup>-3</sup> )                    |  | $5.9 \times 10^{22}$            | $5.9 \times 10^{22}$                                   | $\sim 10^{22}$                        |
| Sample volume (cm <sup>3</sup> )                    |  | 1.2                             | 1.2                                                    | $\sim 1$                              |
| Polarization technique                              |  | Thermal                         | BF <sup>c</sup> / PHIP <sup>d</sup> / DNP <sup>e</sup> | SEOP <sup>f</sup>                     |
| Polarization                                        |  | $1.8 \times 10^{-7}$            | $> 1.1 \times 10^{-5}$                                 | $\sim 1$                              |
| $T_2$ (s)                                           |  | 0.72                            | 1                                                      | 1000                                  |
| $B_0$ inhomogeneity (ppm)                           |  | 10                              | $\leq 2$                                               | $\leq 2$                              |
| Electronic noise ( $\mu\Phi_0/\text{Hz}^{1/2}$ )    |  | 7.1                             | 1.0                                                    | 1.0                                   |
| Flux of SPN <sup>g</sup> ( $\mu\Phi_0$ )            |  | 0.39                            | 0.39                                                   | 0.16                                  |
| Scan range                                          |  | 1.348 450 MHz–<br>1.348 690 MHz | $\sim 1$ kHz–4.3 MHz                                   | $\sim 1$ kHz–1.2 MHz                  |
| $ g_{\text{aNN}} $ sensitivity (GeV <sup>-1</sup> ) |  | $3 \times 10^{-2}$              | $1.0 \times 10^{-6}$ –<br>$1.2 \times 10^{-5}$         | $10^{-12}$ – $10^{-11}$               |

<sup>a</sup> Methanol at 180 K prepolarized at 2 T; measurement time  $100 \tau_a$  per scan step.<sup>b</sup> <sup>129</sup>Xe-enriched liquid sample.<sup>c</sup> Brute-force prepolarization in a strong field at low temperature.<sup>d</sup> Parahydrogen-induced polarization (PHIP).<sup>e</sup> Dynamic nuclear polarization (DNP).<sup>f</sup> Spin-exchange optical pumping (SEOP).<sup>g</sup> Spin-projection noise (SPN).

# 4

## Measurement and Results

Sections 4.1–4.3 are adapted from Ref. [28]. Sections 4.4–4.6 are adapted from Ref. [25].

With the signal model established in the previous chapter, this chapter derives the sensitivity of a single-frequency measurement and then assembles these results into an optimal strategy for scanning the target frequency band.

### 4.1 Single-frequency sensitivity

The transverse magnetization  $M_1$  induced by the ALP field couples to the pickup coil and SQUID via the pickup transfer coefficient  $\alpha$ :

$$\alpha = \frac{\Phi}{\mu_0 M_1}, \quad (4.1)$$

where  $\Phi$  is the magnetic flux in the SQUID and  $\mu_0$  is the vacuum permeability. Using Equations (2.9) and (2.11), the average ALP-induced flux power is

$$\langle \Phi_a^2 \rangle = \frac{1}{2} (\alpha u_n \mu_0 M_0 \Omega_a)^2 \begin{cases} \tau_a T_2^*, & \tau_a \ll T_2^* \\ \frac{\tau_a^2}{\pi}, & T_2^* \ll \tau_a \ll T_2 \\ T_2^2, & T_2 \ll \tau_a \end{cases}. \quad (4.2)$$

The signal amplitude in the power spectral density (PSD) is

$$S = \frac{\langle \Phi_a^2 \rangle}{\Gamma/2\pi}, \quad (4.3)$$

where  $\Gamma$  is the expected signal linewidth. Depending on which linewidth is narrower,

$$\Gamma = \begin{cases} \Gamma_a, & \Gamma_a \ll \Gamma_n \\ \Gamma_n, & \Gamma_a \gg \Gamma_n \end{cases}. \quad (4.4)$$

Within a spectral bin of width  $\Gamma$ , the number of PSD values available for averaging from a measurement of dwell time  $T$  is

$$N_\Gamma = \frac{\Gamma T}{2\pi} \gg 1, \quad (4.5)$$

and the noise standard deviation after averaging is

$$\sigma = \sqrt{\frac{2\pi}{\Gamma T}} \sigma_0, \quad (4.6)$$

where  $\sigma_0$  is the single-bin noise level of the SQUID system.

Using a detection threshold of  $S \geq 3.355 \sigma$ —corresponding to a true signal  $5 \sigma$  above the noise detected at 95% confidence [4]—the ALP coupling strengths excluded when no signal is found satisfy

$$|g_{\text{aNN}}| \geq \begin{cases} \eta T^{-1/4} (T_2^*)^{-3/4} \tau_a^{-1/2}, & \tau_a \ll T_2^* \\ \eta \pi^{-1/4} T^{-1/4} (T_2^*)^{-1} \tau_a^{-1/4}, & T_2^* \ll \tau_a \ll T_2 \\ \eta \pi^{-3/4} T^{-1/4} T_2^{-1} (T_2^*)^{-1} \tau_a^{3/4}, & T_2^* \ll T_2 \ll \tau_a \\ \eta \pi^{-3/4} T^{-1/4} T_2^{-2} \tau_a^{3/4}, & T_2^* = T_2 \ll \tau_a \end{cases}, \quad (4.7)$$

where the prefactor  $\eta$  collects the apparatus-dependent constants:

$$\eta = \frac{2\sqrt{3.355} \sigma_0}{\pi^{1/4} \alpha \mu_0 M_0 v_\perp \sqrt{\hbar c \rho_{DM}}}. \quad (4.8)$$

Larger magnetization  $M_0$  and lower noise  $\sigma_0$  both improve  $\eta$  and hence the sensitivity.

## 4.2 Scanning procedure

The CASPER-Gradient-Low-Field experiment scans the ALP-mass parameter space by sweeping the magnitude of  $\mathbf{B}_0$ , which tunes the Larmor frequency of the sample to a series of values  $\nu_i \in [\nu_{\text{start}}, \nu_{\text{end}}]$ . One step of the scan consists of:

- A. Pulsed-NMR measurement to determine the resonance frequency and the effective linewidth ( $T_2^*$ ).

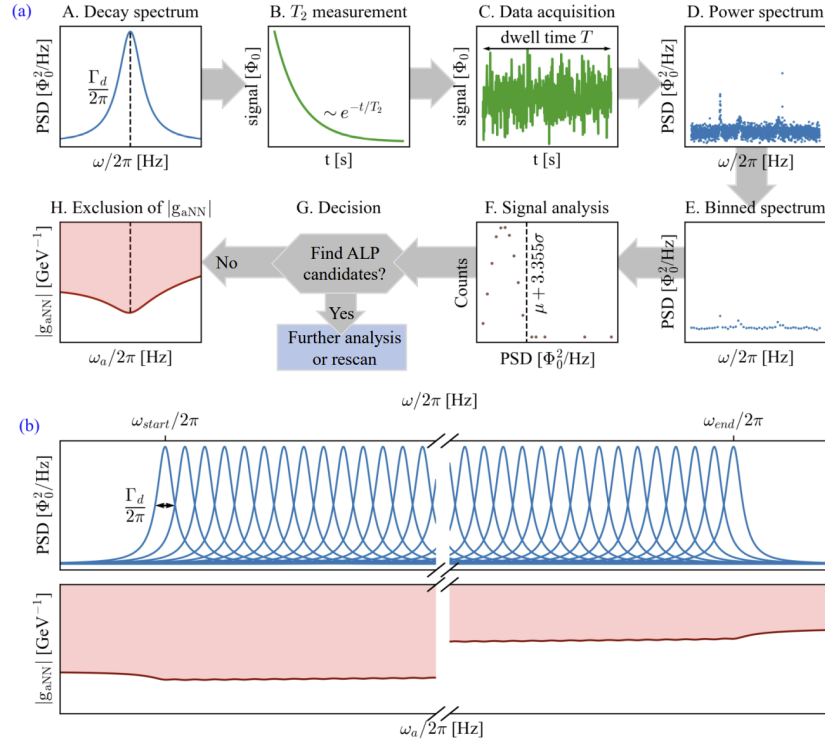
- B. Spin-echo measurement to determine  $T_2$ .
- C. Free-precession data acquisition with the SQUID magnetometer—no transverse pulse applied—to record the ALP-signal channel over the dwell time  $T$ .
- D. Fourier transform of the magnetometer data to produce a power spectrum.
- E. Binning of PSD values within frequency bins of width  $\Gamma$ , and averaging to reduce noise.
- F. Statistical analysis of binned PSD values (e.g., histogram comparison with the expected noise distribution) to identify DM-candidate frequencies.
- G. If a candidate is found: repeat the measurement at the same field value and perform additional checks.
- H. If no candidate is found: record the exclusion contour for this frequency step and ramp  $B_0$  to the next value.

The procedure is illustrated in Figure 4.1. A significant excess surviving repeated measurement can be tested further using the daily and annual modulation signatures of ALP dark matter [15].

The frequency-step size  $\Gamma_{\text{scan}}$  is set by the sensitive bandwidth of a single step. An experiment is sensitive to signals within a bandwidth equal to the maximum of  $\Gamma_n$  and  $\Gamma_a$  around a given Larmor frequency. Choosing a step size larger than  $\Gamma_{\text{scan}}$  leaves gaps; choosing a smaller step size multiplies the number of steps without improving coverage.

## 4.3 Scanning optimization

With fixed total measurement time  $T_{\text{tot}}$ , the dwell time can in principle depend on frequency:  $T \propto \nu^n$ . Maximizing the sensitive area in the  $\log \nu$ – $\log |g_{\text{aNN}}|$  parameter space over the full scan determines the optimal exponent  $n$ . The derivation is given in Appendix C; the result is  $n = 0$  (uniform dwell time) for regimes (1)–(3) and  $n = -1$  (dwell time decreasing as  $1/\nu$ ) for regime (4).



**Figure 4.1:** Schematic of the experimental procedure. (a) One step of the scan: pulsed-NMR diagnostics, free-precession data acquisition, PSD computation and binning, and statistical analysis using a histogram of PSD values. (b) Example decay spectra (blue curves) and the resulting sensitivity region in the ALP mass-coupling space (reddish area) after scanning over a frequency range. Reproduced from Ref. [28].

With optimal dwell-time allocation, the scanning sensitivity is

$$|g_{aNN}| \geq \begin{cases} \eta \left( \frac{T_{\text{tot}}}{Q_a \ln(v_{\text{end}}/v_{\text{start}})} \right)^{-1/4} \times (T_2^*)^{-3/4} \tau_a^{-1/2}, & \tau_a \ll T_2^* \text{ or } T_2^* \ll \tau_a \ll T_2 \\ \eta \left( \frac{\pi^2 T_{\text{tot}}}{Q_a \ln(v_{\text{end}}/v_{\text{start}})} \right)^{-1/4} \times (T_2^*)^{-3/4} T_2^{-1} \tau_a^{1/2}, & T_2^* \ll T_2 \ll \tau_a \text{ or } T_2^* = T_2 \ll \tau_a \end{cases} \quad (4.9)$$

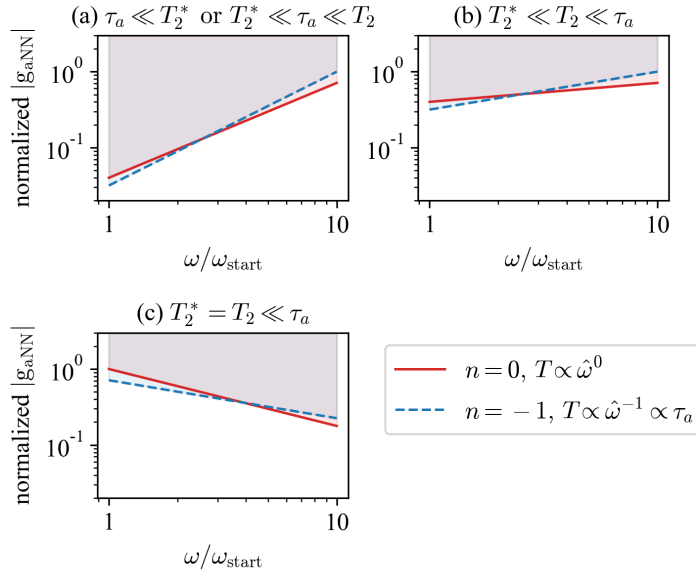
Several conclusions follow directly:

- **Time scaling.** The sensitivity scales as  $T_{\text{tot}}^{-1/4}$  in all regimes, a consequence of the long-dwell-time limit  $T \gg \tau_a, T_2^*, T_2$  [5, 7]. An order-of-magnitude

increase in measurement time improves the exclusion bound by only a factor  $10^{1/4} \approx 1.8$ .

- **Effect of  $T_2^*$ .** Longer  $T_2^*$  is beneficial in all regimes. When  $\tau_a \ll T_2^*$ , it increases the r.m.s. tipping angle and narrows the signal linewidth. When  $T_2^* \ll \tau_a$ , it increases the fraction  $u_n$  of spins on resonance.
- **Effect of  $T_2$ .** Longer  $T_2$  improves sensitivity in the regime  $T_2 \ll \tau_a$ , where  $T_2$  directly limits the r.m.s. tipping angle.

The sensitivity plots computed with uniform ( $n = 0$ ) and frequency-adaptive ( $n = -1$ ) dwell-time allocation for an example scan range  $\nu_{\text{end}}/\nu_{\text{start}} = 10$  are shown in Figure 4.2. The area covered in the  $\log \nu$ – $\log |g_{\text{aNN}}|$  plane is larger with optimal  $n$ .



**Figure 4.2:** Sensitivity plots on a log-log scale for two dwell-time allocation schemes: uniform ( $n = 0$ ) and frequency-adaptive ( $n = -1$ ). The shaded regions indicate the parameter space accessible to the experiment. The coupling strength is normalised in each regime so that the highest point equals 1. The scan range is  $\nu_{\text{end}}/\nu_{\text{start}} = 10$ . Reproduced from Ref. [28].

In summary, the optimal strategy for a frequency-scanning ALP search is to work at the maximum achievable  $T_2$  and  $T_2^*$ , use a step size equal to the sensitive bandwidth, and distribute the total measurement time uniformly across frequency steps (or as  $T \propto \nu^{-1}$  in regime (4)).

## 4.4 The methanol measurement

The scanning framework above was applied in a proof-of-concept ALP search using the CASPER-Gradient-LF apparatus with a thermally polarized methanol sample. This section describes the measurement procedure and parameters. In the following,  $g_{\text{ap}}$  denotes the ALP-proton gradient coupling constant, i.e.  $g_{\text{aNN}}$  evaluated for proton spins.

### 4.4.1 Measurement parameters

The search covered a 240 Hz band centered at 1.348 570 MHz, corresponding to proton Larmor frequencies driven by a bias field  $|\mathbf{B}_0| \approx 31.67$  mT. This maps to ALP masses in the range 5.576741–5.577733 neV/ $c^2$ . The frequency step size was 6 Hz, approximately half the NMR linewidth, so that the resonant windows of adjacent steps overlap. Three sequential scans over the same range were performed. The most sensitive scan was treated as the primary scan and the other two (scans A and B) as consistency checks.

The ALP coherence time at the search frequency is  $\tau_a \approx 0.35$  s, obtained from Equation (2.4). The measured relaxation times are  $T_2^* = 24.2(4.0)$  ms and  $T_2 = 0.72(5)$  s. Since  $T_2^* \ll \tau_a \lesssim T_2$ , the measurement falls in regime (2) of Table 2.1. The spectral factor is approximately  $u \approx T_2^*/\tau_a \approx 7\%$ .

### 4.4.2 Step sequence

At each frequency step the following actions were performed in order:

1. Pulsed-NMR measurement (a  $\pi/2$  pulse followed by free-induction decay) to determine the Larmor frequency, linewidth  $\Delta\nu$  (and hence  $T_2^* = 1/(\pi\Delta\nu)$ ), and NMR signal power. A Lorentzian fit to the resulting PSD peak extracts these parameters and calibrates the sensitivity at each step.
2. Carr-Purcell-Meiboom-Gill (CPMG) [8] pulse sequence prior to the first scan step to determine  $T_2$  from the exponential decay of spin-echo amplitudes.
3. 5 s dwell time to allow ring-down noise to decay and the magnetization to return close to its equilibrium value along  $\mathbf{B}_0$ .
4. 100 s data acquisition without any pulses, recording the SQUID feedback voltage  $V_f$  at 13.4 kHz sampling rate. This window corresponds to approximately 280 ALP coherence times and is the ALP-search channel.

5. Frequency step: the Larmor frequency is increased by  $\approx 6$  Hz by ramping the Helmholtz coils or the superconducting magnet.
6. Another 5 s dwell time after the field ramp before repeating.

The PSDs were recorded after demodulation of  $V_f$  by the lock-in amplifier; its reference frequency was detuned by 820 Hz from the Larmor frequency so that the NMR signal appears at a non-zero difference frequency, making it easily distinguishable from noise. Example PSD spectra and NMR characterisation plots are shown in Figure 4.3.

Due to Earth's rotation, the component of the ALP-field gradient perpendicular to  $\mathbf{B}_0$  varies over the course of the measurement. The resulting daily modulation of the signal amplitude is predictable and was tracked over the approximately six-hour duration of the scans.

## 4.5 Data analysis

Each 100 s time series is converted to a PSD and processed through three successive steps.

### 4.5.1 Spike suppression

A narrow Savitzky-Golay (SG) filter with a window half-width equal to half the expected ALP linewidth at the scanned frequency is applied to the PSD. The residual between the raw PSD and the filter output is fitted with a  $\chi^2$  distribution with two degrees of freedom. Bins that deviate from the filter response by more than the fitted threshold, together with their four nearest neighbours, are replaced by the local filter average. This step suppresses narrow EMI spikes while preserving potential ALP signals. In the primary scan, only 0.002 % of bins were modified in this way.

### 4.5.2 Baseline removal

A reduced chi-square test on the spike-corrected spectrum reveals a non-flat baseline. A second, broad SG filter with a window of 200 ALP linewidths is applied to estimate this baseline. The spike-corrected spectrum is then normalised by subtracting and dividing by this baseline, producing a spectrum with a flat expected noise floor.

### 4.5.3 Matched filtering and NPE

A matched filter using the expected ALP signal lineshape  $\lambda_{\perp}(\nu, \nu_a)$ —the normalised spectral lineshape of the ALP-field gradient component perpendicular to  $\mathbf{B}_0$ , as given in Ref. [15]—is convolved with the normalised PSD. The result, normalised by its standard deviation, is the normalized power excess (NPE). The matched-filter step size is half the ALP linewidth.

### 4.5.4 Hypothesis test and rescan threshold

Frequency bins where the NPE exceeds the rescan threshold  $\Theta_{\text{RE}}$  are flagged as outliers. The threshold was determined via Monte Carlo simulations of the NPE distribution under the null hypothesis that a  $5\sigma$  ALP signal is present, yielding  $\Theta_{\text{RE}} = 3.43$  at the 95 % confidence level (corresponding to a  $p$ -value of 0.001).

The overall analysis efficiency, accounting for the effect of baseline removal on signal recovery, was determined from simulations to be  $\eta_{\text{rec}} = 88.2(4.6)\%$ . This efficiency is applied when converting NPE values to coupling-constant limits.

## 4.6 Exclusion results

### 4.6.1 Candidates and consistency checks

Table 4.1 lists the outlier counts and ALP signal candidates from all three scans. The total number of outliers in each scan exceeds the expectation for white noise, owing to non-white-noise structures that are neither narrow nor broad enough to be removed by the SG filters. By comparing the resonant window (within  $\pm 5$  NMR linewidths of the Larmor frequency) with off-resonance background, and by cross-checking the primary scan against scans A and B, these noise-induced outliers can be identified.

No candidate from the primary scan was found in either of the consistency-check scans within four ALP linewidths. The search therefore yields no statistically significant evidence for ALP dark matter at the detection sensitivity.

### 4.6.2 Coupling-constant limit

In the absence of a detection, an upper limit on the ALP-proton gradient coupling is obtained from

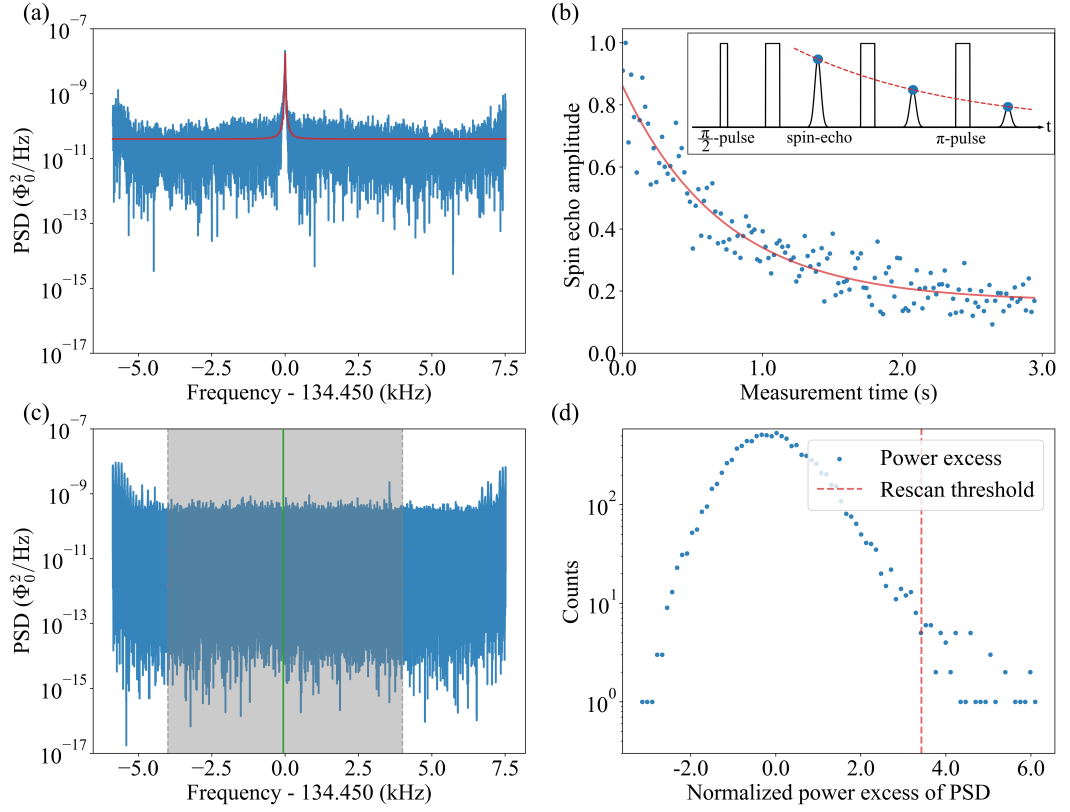
$$|g_{\text{ap, lim}}| = \sqrt{\frac{e_5}{\eta_{\text{rec}} P_{\perp,1}}}, \quad (4.10)$$

**Table 4.1:** Results of the three ALP scans.  $N_{\text{full}}$ : total NPE bins.  $X_{\text{full}}$ : outliers above rescan threshold.  $\langle X_{\text{full}} \rangle$ : expected value assuming white noise.  $N_{\text{res}}$ : bins in resonant windows.  $X_{\text{res}}$ : ALP signal candidates.  $\langle X_{\text{res}} \rangle$ : expected value.  $X'_{\text{res}}$ : candidates after merging those within one ALP linewidth.  $X''_{\text{res}}$ : candidates in this scan also appearing in the primary scan within four ALP linewidths. Parenthetical fractions give the fraction of  $N_{\text{full}}$ .

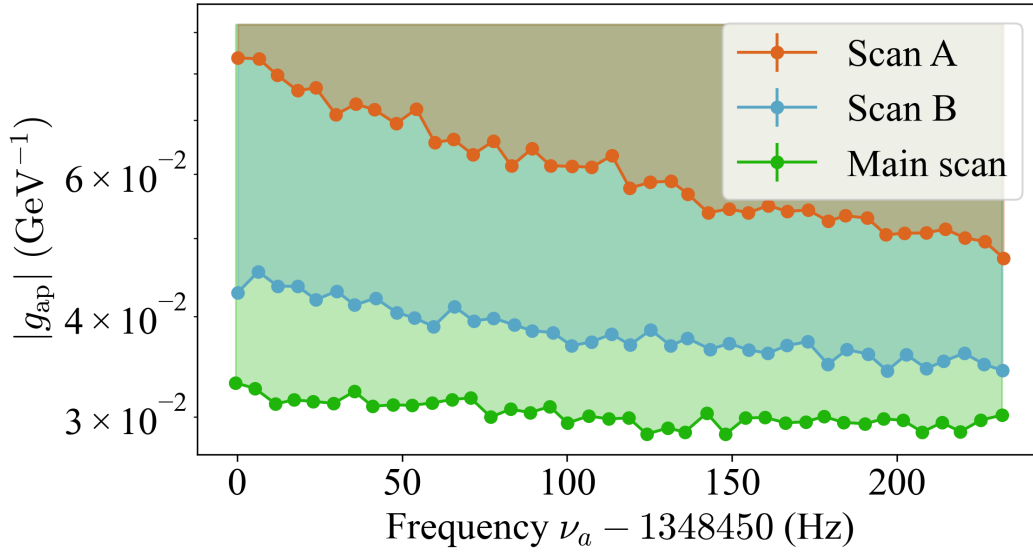
|                                    | Primary scan  | Scan A        | Scan B        |
|------------------------------------|---------------|---------------|---------------|
| $N_{\text{full}}$                  | 481 680       | 481 680       | 481 680       |
| $X_{\text{full}}$                  | 6393 (1.32 %) | 5760 (1.20 %) | 6202 (1.29 %) |
| $\langle X_{\text{full}} \rangle$  | 145 (0.03 %)  | 145 (0.03 %)  | 145 (0.03 %)  |
| $N_{\text{res}}$                   | 8430          | 8430          | 8430          |
| $X_{\text{res}}$                   | 5             | 7             | 3             |
| $\langle X_{\text{res}} \rangle$   | 3             | 3             | 3             |
| $X'_{\text{res}}$                  | 5             | 5             | 3             |
| $X''_{\text{res}}$                 | —             | 0             | 0             |
| $\langle X''_{\text{res}} \rangle$ | —             | 0             | 0             |

where  $e_5$  is the signal power corresponding to the  $5\sigma$  NPE threshold,  $P_{\perp,1}$  is the reference signal power for  $|g_{\text{ap}}| = 1 \text{ GeV}^{-1}$ , and  $\eta_{\text{rec}}$  is the analysis efficiency defined above. The resulting exclusion limits are plotted in Figure 4.4.

The measurement sets the limit  $|g_{\text{ap}}| \lesssim 3 \times 10^{-2} \text{ GeV}^{-1}$  over the scanned ALP mass range  $5.576741\text{--}5.577733 \text{ neV}/c^2$ . This is the first dedicated laboratory search in this mass range. The sensitivity is dominated by the small thermal polarization of the methanol sample; hyperpolarized samples are expected to improve it by several orders of magnitude.



**Figure 4.3:** (a) PSD after a  $\pi/2$  pulse. The NMR peak is fitted with a Lorentzian (red line); the center is at 1 348 449.21(5) Hz with a FWHM of 14.79(1) Hz. (b) CPMG spin-echo amplitudes (blue dots) and exponential fit (red), giving  $T_2 = 0.72(5)$  s. (c) PSD of a 100 s ALP-search step. The grey analysis window (8 kHz wide) excludes filter artefacts near the band edges; the green resonant window ( $\pm 5$  NMR linewidths =  $\pm 70$  Hz) is where ALP signal candidates are sought. (d) Histogram of normalized power excess (NPE) derived from (c), with the rescan threshold  $\Theta_{\text{RE}}$  (red line) at the 95 % confidence level for a  $5\sigma$  ALP detection. Reproduced from Ref. [25].



**Figure 4.4:** ALP gradient-proton coupling excluded at 95 % confidence level by the three sequential scans. Each data point represents the sensitivity for one scan step. The variation in sensitivity between scans reflects the daily modulation of the ALP wind direction relative to the bias field. Reproduced from Ref. [25].



## 5.1 Summary of results

This thesis investigated the use of nuclear magnetic resonance for searches for axionlike dark-matter particles through the nuclear-gradient coupling, focusing on the CASPER-Gradient-Low-Field experiment at Mainz.

The theoretical analysis in Chapter 2 and in Sections 4.1–4.3 of Chapter 4 (adapted from Ref. [28]) established the following key results:

- **Optimal step size.** The frequency-step size for scanning should equal the sensitive bandwidth of a single measurement, namely the larger of the NMR linewidth  $\Gamma_n$  and the ALP linewidth  $\Gamma_a$ .
- **Optimal relaxation times.** The transverse relaxation times  $T_2$  and  $T_2^*$  should be maximised. Longer  $T_2^*$  is beneficial in all regimes: it increases either the r.m.s. tipping angle or the fraction of on-resonance spins. Longer  $T_2$  improves sensitivity when  $T_2 \ll \tau_a$ .
- **Sensitivity scaling.** The sensitivity to  $|g_{aNN}|$  scales as  $T_{\text{tot}}^{-1/4}$  with total measurement time in all regimes considered ( $T \gg \tau_a, T_2^*, T_2$ ). This slow scaling means improving  $T_2, T_2^*$ , or the detector noise  $\sigma_0$  is more efficient than simply accumulating more measurement time.
- **Optimal dwell-time allocation.** For regimes (1)–(3), the dwell time should be distributed uniformly across frequency steps ( $T \propto \nu^0$ ). For regime (4) ( $T_2^* = T_2 \ll \tau_a$ ), the optimal strategy is  $T \propto \nu^{-1}$ .
- **Four regimes.** The parameter space is divided into four regimes by the relative magnitudes of  $T_2, T_2^*$ , and  $\tau_a$ . Each regime has a distinct signal amplitude and optimal dwell time, as summarised in Table 2.1.

These results were applied in a proof-of-principle ALP search with the CASPER-Gradient-LF apparatus (adapted from Ref. [25]), described in Sections 4.4–4.6 of Chapter 4:

- **First laboratory search in the target mass range.** A thermally polarized methanol sample was used to search for the ALP-proton gradient coupling in the mass range  $5.576741\text{--}5.577733\text{ neV}/c^2$  (Compton frequency 240 Hz around 1.348 570 MHz). This is the first dedicated laboratory search in this mass range.
- **No dark-matter candidates found.** Three sequential scans were performed. No ALP signal candidates survived the cross-scan consistency check, yielding the constraint  $|g_{\text{ap}}| \lesssim 3 \times 10^{-2} \text{ GeV}^{-1}$  at 95 % confidence level.
- **Methodology validated.** The data-analysis pipeline—Savitzky-Golay spike suppression, baseline removal, matched filtering, and NPE hypothesis test—was validated through synthetic-signal injection. The overall analysis efficiency is  $\eta_{\text{rec}} = 88.2(4.6) \%$ .
- **Sensitivity limited by thermal polarization.** The proton thermal polarization at 180 K and 31.67 mT is only  $\approx 1.8 \times 10^{-7}$ , which dominates the sensitivity budget. The apparatus is otherwise ready for more sensitive searches with hyperpolarized samples.

## 5.2 Outlook

The framework developed here can be extended in several directions:

- **Hyperpolarized samples.** Hyperpolarized  $^{129}\text{Xe}$  or methanol can increase the polarization by six or more orders of magnitude relative to thermal polarization, directly improving  $M_0$  and hence the sensitivity prefactor  $\eta$  in Equation (4.8). The  $T_1$  relaxation of hyperpolarized samples limits the measurement window and modifies the optimal dwell-time strategy. Examples of projected sensitivities are given in Table 3.2.
- **Seeded measurements.** Applying a transverse oscillating field at the resonance frequency (“seeding”) to heterodyne the ALP signal was outside the scope of this work. Seeding could improve sensitivity by coherently amplifying the ALP-induced tipping angle [5].
- **Improved field homogeneity and longer  $T_2$ .** Reaching the design inhomogeneity of 2 ppm over the sample volume would lengthen  $T_2^*$  and increase the fraction of on-resonance spins. Combined with samples having long intrinsic  $T_2$ , this improvement is beneficial in all four regimes.

- **High-field extension.** The CASPER-Gradient-High-Field experiment will operate with a 14.1 T magnet, reaching Larmor frequencies up to 600 MHz and accessing higher ALP masses. The scanning framework derived here applies directly to that regime.
- **Low-frequency regime.** Below about 1 kHz,  $\tau_a$  can exceed 1000 s, and the assumption  $T \gg \tau_a$  breaks down. An adaptive scanning strategy that accounts for the long coherence time will be required for CASPER-ZULF [13] and related low-frequency experiments.



*This appendix is adapted from Ref. [27].*

Analytical estimates of the ALP-induced spin-precession signal (Chapter 2) are valid in the long-dwell-time limit  $T \gg \tau_a, T_2^*, T_2$ . At lower Compton frequencies, the ALP coherence time becomes comparable to or longer than experimentally accessible dwell times, and the scaling laws no longer apply. Numerical simulation is then the tool of choice. This appendix describes `axionbloch`, an open-source Python package that simulates spin dynamics under both magnetic and axion-induced pseudomagnetic fields.

## A.1 Spin dynamics: the Bloch equations

The package models spin-1/2 ensembles through the Bloch equations. In the presence of a total effective magnetic field  $\mathbf{B}_{\text{eff}}$  (which includes both the laboratory bias field and the axion-induced pseudomagnetic field), the magnetization  $\mathbf{M}$  evolves as

$$\frac{dM_x}{dt} = \gamma(\mathbf{M} \times \mathbf{B}_{\text{eff}})_x - \frac{M_x}{T_2}, \quad (\text{A.1})$$

$$\frac{dM_y}{dt} = \gamma(\mathbf{M} \times \mathbf{B}_{\text{eff}})_y - \frac{M_y}{T_2}, \quad (\text{A.2})$$

$$\frac{dM_z}{dt} = \gamma(\mathbf{M} \times \mathbf{B}_{\text{eff}})_z - \frac{M_z - M_0}{T_1}, \quad (\text{A.3})$$

where  $\gamma$  is the gyromagnetic ratio and  $M_0$  is the equilibrium magnetization. The bias field is usually much stronger than the pseudomagnetic field, so the equations are solved in the rotating coordinate frame (RCF), which removes the fast Larmor precession and exposes the slow axion-induced dynamics. The effective transverse relaxation time  $T_2^*$  (which includes inhomogeneous broadening) is simulated by sampling the spread of bias-field values from a `Magnet` instance and integrating the Bloch equations for each sampled field independently.

## A.2 Numerical integration

The time evolution of  $\mathbf{M}$  is obtained by numerically integrating Equations (A.1–A.3) with the fourth-order Runge–Kutta (RK4) method. Given a time step  $\Delta t$  and initial magnetization  $\mathbf{M}^n$ , the update rule is

$$\mathbf{M}^{n+1} = \mathbf{M}^n + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4), \quad (\text{A.4})$$

where  $\mathbf{k}_1 = d\mathbf{M}/dt|_{\mathbf{M}^n}$  and subsequent  $\mathbf{k}_i$  are evaluated at intermediate half-steps following the standard RK4 prescription. The local truncation error is  $\mathcal{O}(\Delta t^5)$  and the global accumulated error is  $\mathcal{O}(\Delta t^4)$ . The time step is set to be at least one order of magnitude smaller than the shortest characteristic timescale in the simulation (e.g.,  $T_1$ ,  $T_2$ , or  $1/\nu_a$ ).

The RK4 integration is implemented in a compiled C++ backend and exposed to Python through the `pybind11` interface, achieving runtimes of  $\sim 10$  s for  $\approx 4 \times 10^9$  integration steps on a modern desktop CPU.

## A.3 Software architecture

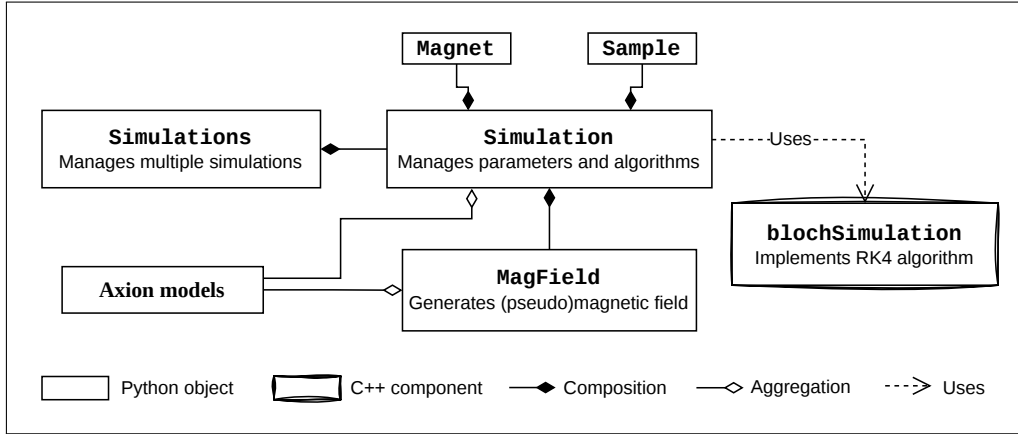
The package uses an object-oriented design separating three concerns: experimental configuration (`Sample`, `Magnet`), field generation (`MagField`), and simulation management (`Simulation`, `Simulations`). Axion-field models are encapsulated in dedicated classes (e.g., `MilkyWayAxionHalo`), allowing new models to be added without modifying the simulation framework. The architecture is illustrated in Figure A.1.

## A.4 Calibration

Three standard NMR experiments are used to benchmark the simulation against known analytical results.

### A.4.1 Pulsed NMR: free decay and spin echo

After a  $90^\circ$  pulse, the magnetization tips into the transverse plane and decays with the dephasing time  $T_2^*$  (free-induction decay). A subsequent  $180^\circ$  pulse refocuses the inhomogeneous dephasing, producing a spin echo whose amplitude decays with the true  $T_2$ . The simulated signals reproduce both behaviours, calibrating



**Figure A.1:** Architecture of axionbloch. Each `Simulation` instance receives parameters from `Magnet`, `Sample`, and `MagField`. For axion simulations, an axion-model object is required both to configure the simulation and to generate the pseudomagnetic field. Numerical integration is performed via the `blochSimulation` API, which provides Python access to the C++ backend. Reproduced from Ref. [27].

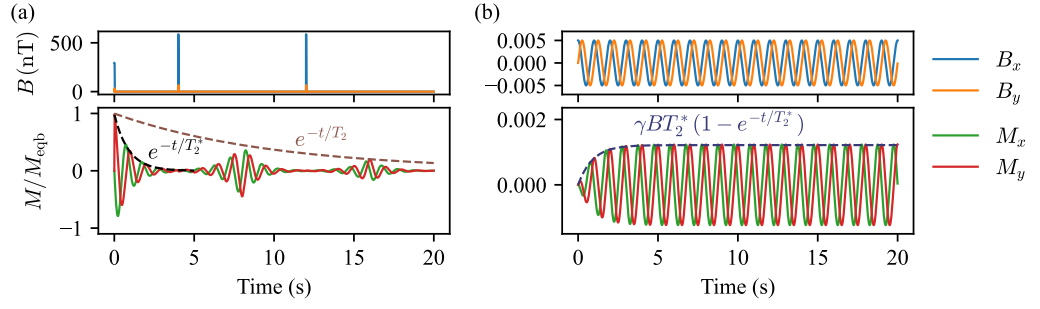
the pulse-induced rotation, Larmor precession, and transverse relaxations. The simulated magnetization is shown in Figure A.2 (a).

### A.4.2 Continuous-wave NMR

For a weak oscillating field on resonance ( $\gamma B T_2^* \ll 1$ ), the transverse magnetization grows as  $\gamma B T_2^* (1 - e^{-t/T_2^*})$  [2]. The simulation reproduces this analytical form, as shown in Figure A.2 (b). This benchmark calibrates both the Larmor precession and the  $T_2^*$  relaxation under continuous driving—the same physical regime as the ALP-induced signal.

### A.4.3 Hyperpolarized sample and $T_1$ relaxation

Hyperpolarization boosts the initial spin polarization above its thermal value. In the simulation, the initial polarization is set as a parameter of the `Sample` class. A sample polarized significantly above thermal equilibrium relaxes toward  $M_0$  with the longitudinal time constant  $T_1$ . The simulated  $M_z(t) = M_0[1 - (M_0^{\text{init}}/M_0 - 1)e^{-t/T_1}]$  reproduces the expected exponential approach to equilibrium, as illustrated in Figure A.3.



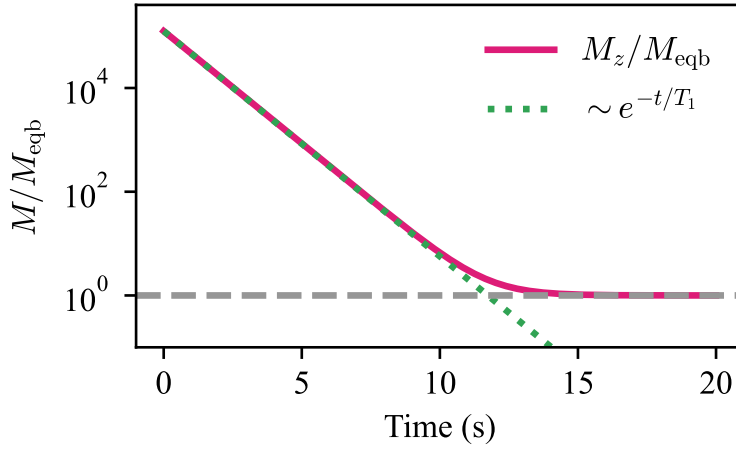
**Figure A.2:** Calibration of axionbloch against analytical NMR results. Upper rows: applied transverse magnetic fields; lower rows: normalized simulated magnetization. **(a)** After  $90^\circ$  and  $180^\circ$  pulses, the simulation produces free-induction decay (timescale  $T_2^*$ ) and a spin echo (timescale  $T_2$ ). **(b)** Under a weak continuous-wave drive on resonance, the transverse magnetization grows as  $\gamma B T_2^* (1 - e^{-t/T_2^*})$  (dashed line). Reproduced from Ref. [27].

## A.5 Axion simulation example

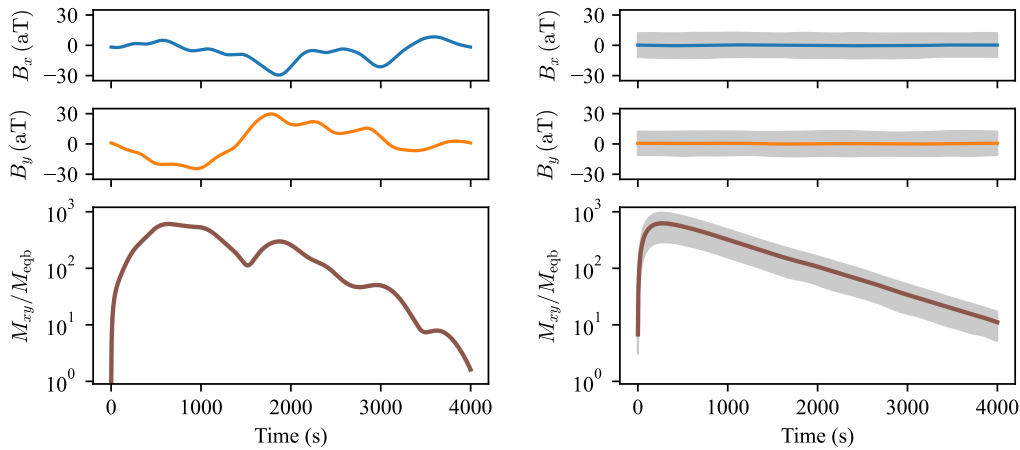
As a demonstration, the MilkyWayAxionHalo model simulates the pseudomagnetic field from virialized galactic dark matter following the standard halo model [12, 15]. The axion field oscillates at the Compton frequency with a Maxwell-Boltzmann velocity distribution setting the spectral linewidth. The stochasticity of the field amplitude is accounted for by generating many independent realisations of the pseudomagnetic field.

Figure A.4 shows results for a  $^{129}\text{Xe}$  sample polarized to 50 %. A single realisation (left) illustrates the stochastic fluctuations in both the pseudomagnetic field and the induced transverse magnetization. An ensemble of 1000 realisations (right) yields the mean signal and its standard deviation band. The signal initially grows as the ALP field drives the magnetization away from equilibrium, and then decays on the  $T_1$  timescale as the hyperpolarization is lost.

These simulations validate the three-regime picture in Equation (2.11): at low frequencies where  $\tau_a \gg T$ , the signal grows linearly in dwell time; in the intermediate regime  $\tau_a \sim T$ , the signal departs from the analytical long-dwell-time scaling; and at high frequencies where  $T \gg \tau_a$ , the analytical results apply directly.



**Figure A.3:** Simulated  $T_1$  relaxation of a hyperpolarized sample in a static bias field. The axial magnetization  $M_z$  is normalized by the equilibrium magnetization  $M_0$ . The exponential approach to equilibrium calibrates the longitudinal relaxation in the simulation. Reproduced from Ref. [27].



**Figure A.4:** Simulated axion-induced NMR signals for a virialized Milky Way axion dark matter halo. Upper panels: pseudomagnetic fields  $B_x$  and  $B_y$  in the transverse plane. Lower panels: amplitude of the induced transverse magnetization  $M_{xy}$ , normalized by  $M_0$ . Left: single simulation realisation. Right: ensemble of 1000 realisations; solid lines show the mean and shaded bands show  $\pm\sigma$ . The sample is  $^{129}\text{Xe}$  initially polarized to 50 %. Reproduced from Ref. [27].



## B.1 The Neyman–Pearson Lemma

The *Neyman–Pearson lemma* is a foundational result in statistical hypothesis testing. It establishes the form of the most powerful test for discriminating between two simple hypotheses at a fixed significance level.

### B.1.1 Problem setup

Suppose we observe data  $x$  drawn from a probability distribution. We wish to test

$$H_0 : x \sim f_0(x)$$

against

$$H_1 : x \sim f_1(x),$$

where both  $f_0(x)$  and  $f_1(x)$  are fully specified probability density functions. The goal is to construct a test that maintains a fixed false-alarm probability (Type I error) while maximising the probability of correctly detecting the alternative hypothesis (power).

### B.1.2 Likelihood ratio and the lemma

The key quantity is the likelihood ratio

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)}.$$

For a fixed significance level  $\alpha$ , the Neyman–Pearson lemma states that the most powerful test rejects  $H_0$  when  $\Lambda(x) > k$ , where  $k$  satisfies  $P(\Lambda(x) > k \mid H_0) = \alpha$ . The likelihood-ratio test is therefore optimal among all tests with the same false-positive rate.

### B.1.3 Type I error and power

The significance level  $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$  is the false-alarm probability. The power  $1 - \beta = P(\text{reject } H_0 \mid H_1 \text{ true})$  is the probability of correctly detecting an ALP signal. The Neyman–Pearson lemma guarantees that the likelihood-ratio test maximises the power for fixed  $\alpha$ .

### B.1.4 Application to the ALP search

In the ALP data analysis described in Section 4.5, the normalized power excess (NPE) plays the role of the test statistic. The null hypothesis is that a  $5\sigma$  ALP signal is present; the alternative is that no signal is present. Monte Carlo simulations of the NPE distribution under the null hypothesis determine the threshold  $\Theta_{\text{RE}} = 3.43$  corresponding to the 95 % confidence level (see Section 4.5).

### B.1.5 Gaussian example

For  $X \sim \mathcal{N}(\mu, \sigma^2)$  with  $H_0 : \mu = 0$  versus  $H_1 : \mu = 1$ , the log-likelihood ratio simplifies to  $\log \Lambda(x) = x/\sigma^2 - 1/(2\sigma^2)$ , so rejecting for large  $\Lambda(x)$  is equivalent to rejecting for large  $x$ . This confirms that the sample mean is the optimal test statistic for Gaussian location testing.

*This appendix is adapted from Ref. [28].*

This appendix derives the optimal dwell-time allocation scheme for a frequency scan, following the treatment of Ref. [28]. The goal is to maximise the sensitive area in the  $\log \nu$ – $\log |g_{\text{aNN}}|$  parameter space over a total measurement time  $T_{\text{tot}}$ .

To simplify the notation, the frequency is normalised:  $\hat{\nu} = \nu/\nu_{\text{start}}$ , with  $\hat{\nu} \in [1, \hat{\nu}_{\text{end}}]$  where  $\hat{\nu}_{\text{end}} = \nu_{\text{end}}/\nu_{\text{start}} > 1$ . The dwell time is parametrised as

$$T = \kappa \hat{\nu}^n, \quad (\text{C.1})$$

where  $\kappa$  is determined by  $T_{\text{tot}}$  and  $n$ , and the optimal value of  $n$  is determined below.

## Regimes (1)–(3): step size proportional to $\nu$

In regimes (1), (2), and (3), the frequency-step size is proportional to  $\hat{\nu}$  (either one ALP linewidth  $\hat{\nu}/Q_a$  or one NMR linewidth  $\delta_\nu \hat{\nu}$ ). The scan rate is

$$\frac{d\hat{\nu}}{dt} = \frac{\Gamma_{\text{scan}}}{T} \propto \hat{\nu}^{1-n}, \quad (\text{C.2})$$

so that

$$dt = \kappa C \hat{\nu}^{n-1} d\hat{\nu}, \quad (\text{C.3})$$

where  $C = Q_a$  in regime (1) and  $C = \delta_\nu^{-1}$  in regimes (2) and (3). Integrating over  $[0, T_{\text{tot}}]$  and  $[1, \hat{\nu}_{\text{end}}]$  gives

$$\kappa = \frac{n T_{\text{tot}}}{C(\hat{\nu}_{\text{end}}^n - 1)}, \quad n \neq 0. \quad (\text{C.4})$$

At  $n = 0$ :  $\kappa|_{n=0} = T_{\text{tot}}/[C \ln(\hat{\nu}_{\text{end}})]$ .

## Regime (4): step size independent of $\nu$

In regime (4) ( $T_2^* = T_2 \ll \tau_a$ ), the step size is one NMR linewidth  $2/T_2$ , independent of frequency. The scan rate is proportional to  $\hat{\nu}^{-n}$ , giving

$$\kappa = \frac{2(n+1) T_{\text{tot}}}{(\hat{\nu}_{\text{end}}^{n+1} - 1) T_2}, \quad n \neq -1. \quad (\text{C.5})$$

At  $n = -1$ :  $\kappa|_{n=-1} = 2T_{\text{tot}}/[\ln(\hat{\nu}_{\text{end}}) T_2]$ .

## Optimal exponent $n$

The sensitive area in log-log scale is

$$A(n) = - \int_1^{\hat{\nu}_{\text{end}}} \log f(\hat{\nu}, n) d \log \hat{\nu}, \quad (\text{C.6})$$

where  $f(\hat{\nu}, n)$  is the exclusion threshold from Equation (4.7) with the dwell time from Equation (C.1). Taking the derivative  $A'(n)$  and setting it to zero shows:

- For regimes (1), (2), and (3):  $A'(n) > 0$  for  $n < 0$  and  $A'(n) < 0$  for  $n > 0$ , so the maximum is at  $n = 0$  (uniform dwell time).
- For regime (4): the analysis shifts the effective exponent from  $n$  to  $n + 1$ , giving an optimum at  $n = -1$  (dwell time  $T \propto \hat{\nu}^{-1}$ ).

An intuitive restatement: when the sensitive bandwidth per step grows with frequency (regimes 1–3), the number of steps per unit  $\log \nu$  is constant, and uniform dwell time is optimal. When the bandwidth is independent of frequency (regime 4), spending less time at higher frequencies—where the ALP coherence time is shorter and the signal integrates faster—maximises the covered area.

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